



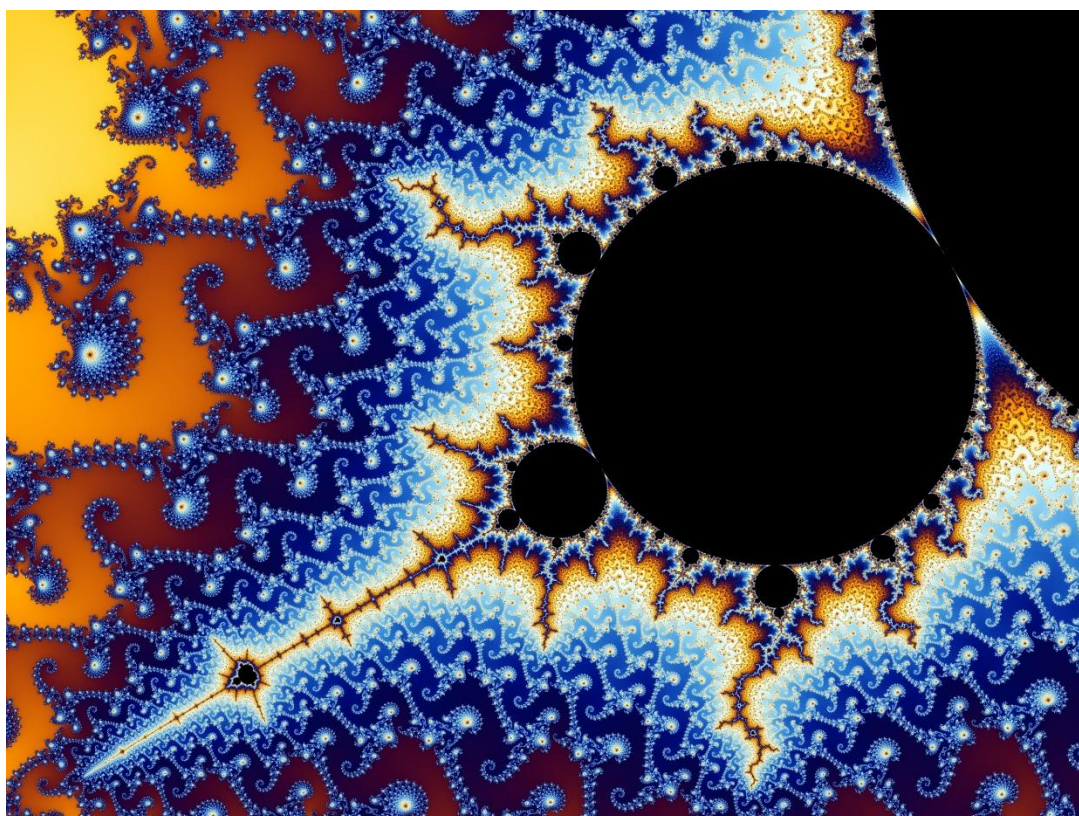
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Tour de Maths 2023

Leg 4

Question Paper - MEMO



The Mandelbrot set

5 Mark Questions

Question 1 How many numbers below one hundred are divisible by both 2 and 3?	MEMO A number that is divisible by 2 and 3 will also be divisible by 6 ANSWER:16
Question 2 Sihle has 27 cards. This is three more than four times the number of cards his brother has. If Sihle's brother has x cards, write down an equation to represents this information?	MEMO ANSWER: $27 = 4x + 3$
Question 3 The ratio of cars to motorbikes in a car park is 10 cars to 4 motorbikes. There are 70 motorbikes in the car park. How many cars are there?	MEMO 10 cars: 4 motorbikes X: 70 motorbikes $X = 2,5 * 70 = 175$ ANSWER:175
Question 4 When I was 14 years old my mother was 42 years old, which was three times my age. Now she is twice my age, how old am I?	MEMO Let x be the father's age and his child's age be y . The father is therefore $x = y + 28$. Today: The father $x = 2y$, hence $2y = y + 28$ and so $y = 28$. ANSWER: 28

Question 5 A tortoise and a hare race against each other. A hare runs at a constant speed of 36 km per hour for exactly ten seconds and waits for the tortoise to catch up. If the tortoise takes two hours to move 1 km, how long will it take to catch up?	MEMO 36 km/h is equivalent to 10 m /s. Hare has thus run 100 m per 10 seconds. The tortoise takes 120 minutes to move 1000 m. It will thus take 12 minutes to move 100 m. ANSWER:12
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Question 6 Two ordinary six-sided dice are stacked on top of each other and placed on a tabletop. What is the sum of the dots on all the visible faces?	MEMO ANSWER: 29
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Question 7 Mr. and Mrs. Dlamini have two daughters and three sons. At Easter time every member of the family buys one chocolate Easter egg for each other member. How many Easter eggs will be bought in total?	MEMO This family has 7 members. Each person buys 6 eggs. ANSWER: 42
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Question 8 The digits 1, 2, 3, 4, and 9 are each used once to form the smallest possible even five-digit number. What is the digit in the tens place?	MEMO The smallest even number is 12394. ANSWER: 9
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Question 9	MEMO
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Three solid balls, each of radius 3,25 cm, are stored in a cylindrical can with the smallest possible radius and volume. What fraction of the can's volume is air?

$$\text{Volume of 3 spheres} = 3 \left[\frac{4}{3} \pi r^3 \right] = 431.28$$

$$\text{Volume of cylinder} = \pi r^2 h = 647.06$$

$$\text{Volume of air: } 647.06 - 431.28 = 215.78 \approx \frac{1}{3} \text{ volume of cylinder}$$

ANSWER: $\frac{1}{3}$

Question 10

Three angles of a pentagon have measure 88° , 124° and 92° . If the measures of the remaining two angles are equal, what is the measure, in degrees, of one of the remaining angles?

MEMO

5 angles measure 540° . The two unknown angles combined measure 236° . Since they are equal, each measures **118°** .

ANSWER: 118°

10 Mark Questions

Question 11 Before he took his last exam, the average of John's exam scores was 89. He figures that if he scores 97 on the last exam, his average for all the exams will be 90. If he scores a 73, his average for all the exams will be 87. How many exams, including the last one, are given in the class?	MEMO Let n be the number of exams written including the last one. $n-1$ exams have already been written. scores for these exams amount to $89(n-1)$ $\frac{89(n-1) + 97}{n} = 90$ $n = 8$ ANSWER: 8
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Question 12 Real numbers a and b satisfy the equations $3^a = 81^{b+2}$ and $125^b = 5^{a-3}$. What is ab?	MEMO $a = 4b + 8$ $3b = a - 3$ $a = -12; b = -5$ ANSWER: 60
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Question 13 The number 13 is prime and so too is its reverse, 31. How many two-digit primes can you find for which their reverse is also prime?	MEMO A prime cannot have an even tens digit since its reverse will be an even number. Similarly, we cannot have 5 tens. Only the following numbers will have reverse primes: 11, 13, 17, 31, 37, 71, 73, 79, 97 ANSWER: 9
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Question 14 Find the value of the digit c in the following calculation. $\begin{array}{r} a\ b \\ -b\ a \\ \hline c\ 4 \end{array}$	MEMO Writing $(10a + b) - (10b + a) = 9(a - b)$, so the difference must be a multiple of 9. The only 2-digit multiple of nine ending with the digit 4 is 54. $C = 5$. ANSWER: 5
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Question 15

Let $a = 1$, $b = 2$ and $c = 3, \dots, z = 26$. What is the exact value of

$$((x - a)(x - b) \dots (x - z))?$$

MEMO

One of the factors will be $(x - x) = 0$. The value of the product will therefore also be a zero.

ANSWER: 0

20 Mark Questions

Question 16

The rule of an obstacle course specifies that at the n th obstacle a person has to toss a die n times. If the sum of points in these tosses is bigger than 2^n , the person is said to have crossed the obstacle. At most how many obstacles can a person cross?

MEMO

For a fair die, the probability of any number appearing is the same.

Since the highest point of a die is 6 and $6 \times 4 > 2^4$, $6 \times 5 < 2^5$, it is impossible that the sum of points appearing in n tosses is bigger than 2^n if $n \geq 5$.

ANSWER: 4

Question 17

Sarah started school at the age of five. She spent one quarter of her life being educated and went straight to work. After working for one half of her life, she lived for fourteen happy years after retiring. How old was she when she retired?

MEMO

75% of Sarah's life is spent studying and working.
25% of her life will include 5 years before school and 14 years after her retirement (a total of 19 years).

$$75\% \text{ of her life} = 19 \times 3 = 62$$

ANSWER: 62

Question 18

A cube measuring $3 \times 3 \times 3$ is made up of 27 smaller cubes and has a 1×1 square cube hole pushed through the centre of each face so that you can see straight through the cube from every side.



MEMO

A solid cube measuring: $5 \times 5 \times 5 = 125$ cubes

$3 \times 3 \times 3 = 27$ cubes is removed from the centre and six faces of $3 \times 3 = 9$ cubes are removed (54).

44 small cubes remain.

ANSWER: 44

The number of small cubes remaining is 20. If a 5 x 5 x 5 cube has 3 x 3 square holes pushed through the centre of each face, how many smaller cubes would remain?

Question 19

In a group of 35 children, 10 have blonde hair, 14 have brown eyes, and 4 have blonde hair and brown eyes. What fraction of the class has blonde hair or brown eyes?

MEMO

$$\frac{10}{35} + \frac{14}{35} - \frac{4}{35} = \frac{20}{35} = \frac{4}{7}$$

ANSWER: $\frac{4}{7}$

Question 20

A train travelling between two towns arrives ten minutes late when travelling at 40 km/h and sixteen minutes late when travelling at 30 km/h. What is the distance between the two towns?

MEMO

Let t be normal time to travel between the two towns.

Distance = speed * time

40 km/h = 0.67 km / minute

30 km/h = 0.5 km / minute

$0.67(t+10) = 0.5(t+16)$

$t \approx 8$

Therefore, distance = $0.67(8+10) = 12$ km

ANSWER: 12

Total: 200

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